

SYSTEM DOCUMENTATION



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**PROGRAM STUDI TEKNIK INFORMATIKA
FAKULTAS ILMU KOMPUTER
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Pendahuluan

Dokumen ini berisi penjelasan lengkap mengenai sistem aplikasi pemesanan makanan kantin berbasis web yang dibuat dengan menggunakan framework Laravel. Isi dokumentasi mencakup perancangan sistem seperti migrasi database, struktur tampilan antarmuka (frontend), penggunaan API, serta pengamanan data. Tujuan dari dokumen ini adalah untuk memberikan gambaran yang jelas mengenai fitur dan cara kerja system.

I. Perancangan Sistem

1. Pembuatan Struktur Migration

```
<?php

use Illuminate\Database\Migrations\Migration;
use Illuminate\Database\Schema\Blueprint;
use Illuminate\Support\Facades\Schema;

return new class extends Migration
{
    /**
     * Run the migrations.
     */
    public function up(): void
    {
        Schema::create('orders', function (Blueprint $table) {
            $table->id();
            $table->foreignId('user_id')->constrained()->onDelete('cascade');
            $table->foreignId('menu_id')->constrained()->onDelete('cascade');
            $table->integer('quantity');
            $table->text('notes')->nullable();
            $table->enum('status', ['pending', 'processing', 'ready', 'done'])->default('pending');
            $table->timestamps();
        });
    }

    /**
     * Reverse the migrations.
     */
    public function down(): void
    {
        Schema::dropIfExists('orders');
    }
};
```

```
<?php

use Illuminate\Database\Migrations\Migration;
use Illuminate\Database\Schema\Blueprint;
use Illuminate\Support\Facades\Schema;

return new class extends Migration
{
    /**
     * Run the migrations.
     */
    public function up(): void
    {
        Schema::create('menus', function (Blueprint $table) {
            $table->id();
            $table->string('name');
            $table->text('description')->nullable();
            $table->integer('price');
            $table->date('available_date');
            $table->timestamps();
        });
    }

    /**
     * Reverse the migrations.
     */
    public function down(): void
    {
        Schema::dropIfExists('menus');
    }
};
```

```

<?php

use Illuminate\Database\Migrations\Migration;
use Illuminate\Database\Schema\Blueprint;
use Illuminate\Support\Facades\Schema;

return new class extends Migration
{
    /**
     * Run the migrations.
     */
    public function up(): void
    {
        Schema::create('notifications', function (Blueprint $table) {
            $table->id();
            $table->foreignId('user_id')->constrained()->onDelete('cascade');
            $table->string('title');
            $table->text('message');
            $table->boolean('read')->default(false);
            $table->timestamps();
        });
    }

    /**
     * Reverse the migrations.
     */
    public function down(): void
    {
        Schema::dropIfExists('notifications');
    }
};

```

```

<?php

use Illuminate\Database\Migrations\Migration;
use Illuminate\Database\Schema\Blueprint;
use Illuminate\Support\Facades\Schema;

return new class extends Migration
{
    /**
     * Run the migrations.
     */
    public function up(): void
    {
        Schema::create('users', function (Blueprint $table) {
            $table->id();
            $table->string('avatar_url')->nullable();
            $table->string('name');
            $table->string('email')->unique();
            $table->timestamp('email_verified_at')->nullable();
            $table->string('password');
            $table->enum('role', ['admin', 'user'])->default('user');
            $table->rememberToken();
            $table->timestamps();
        });

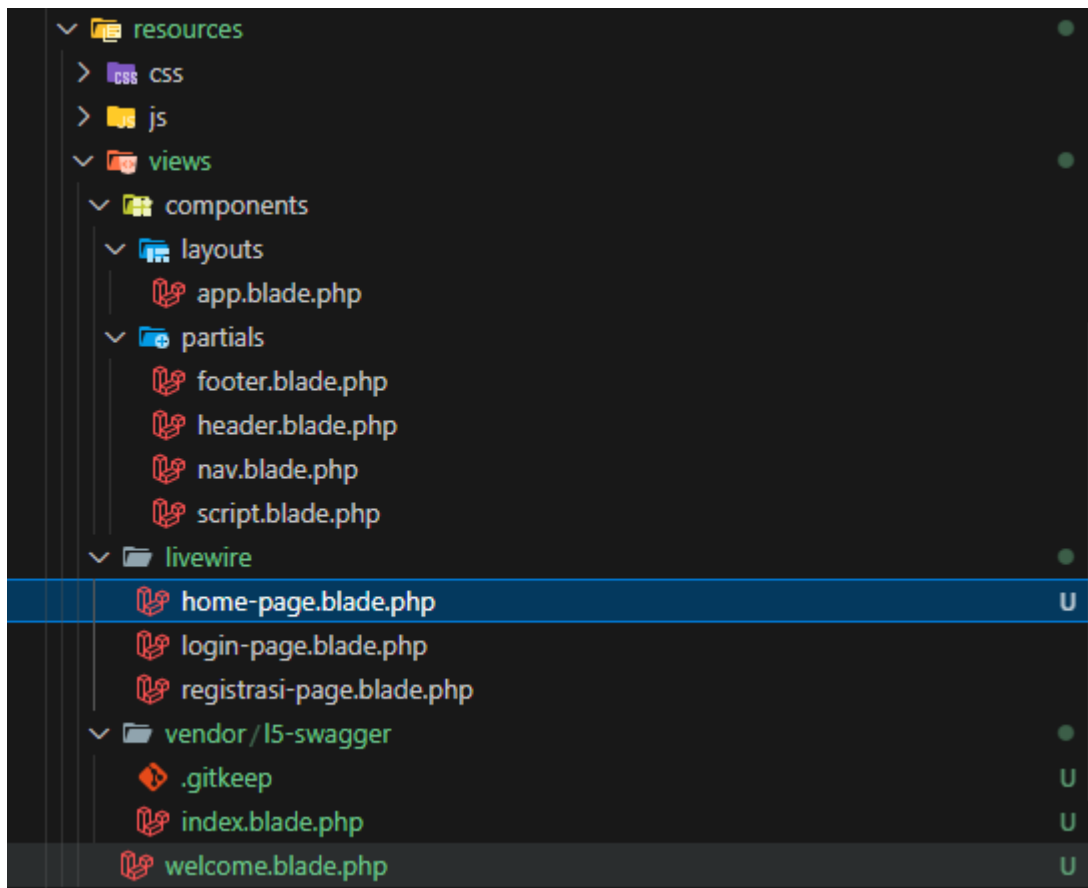
        Schema::create('password_reset_tokens', function (Blueprint $table) {
            $table->string('email')->primary();
            $table->string('token');
            $table->timestamp('created_at')->nullable();
        });

        Schema::create('sessions', function (Blueprint $table) {
            $table->string('id')->primary();
            $table->foreignId('user_id')->nullable()->index();
            $table->string('ip_address', 45)->nullable();
            $table->text('user_agent')->nullable();
            $table->longText('payload');
            $table->integer('last_activity')->index();
        });
    }

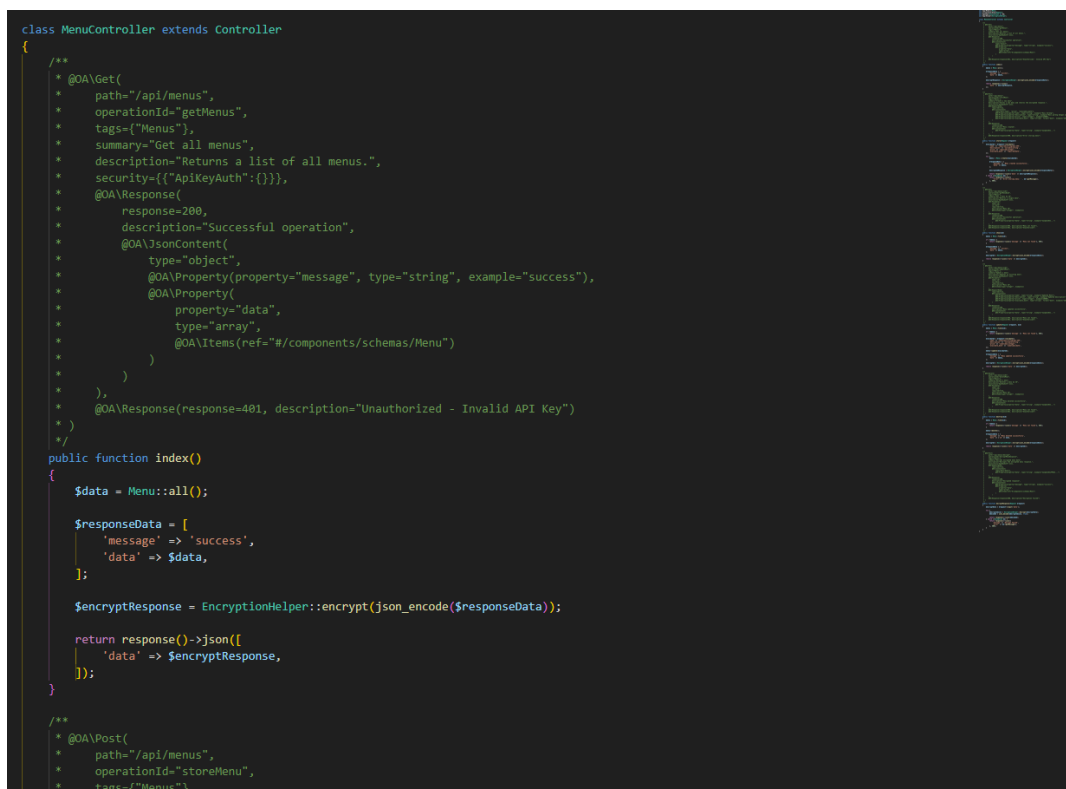
    /**
     * Reverse the migrations.
     */
    public function down(): void
    {
        Schema::dropIfExists('users');
    }
};

```

2. pembuatan Struktur file Frontend



3. Pembuatan Api Controller di masing masing migration



4. Pembuatan Api routes di masing masing migration

```
<?php

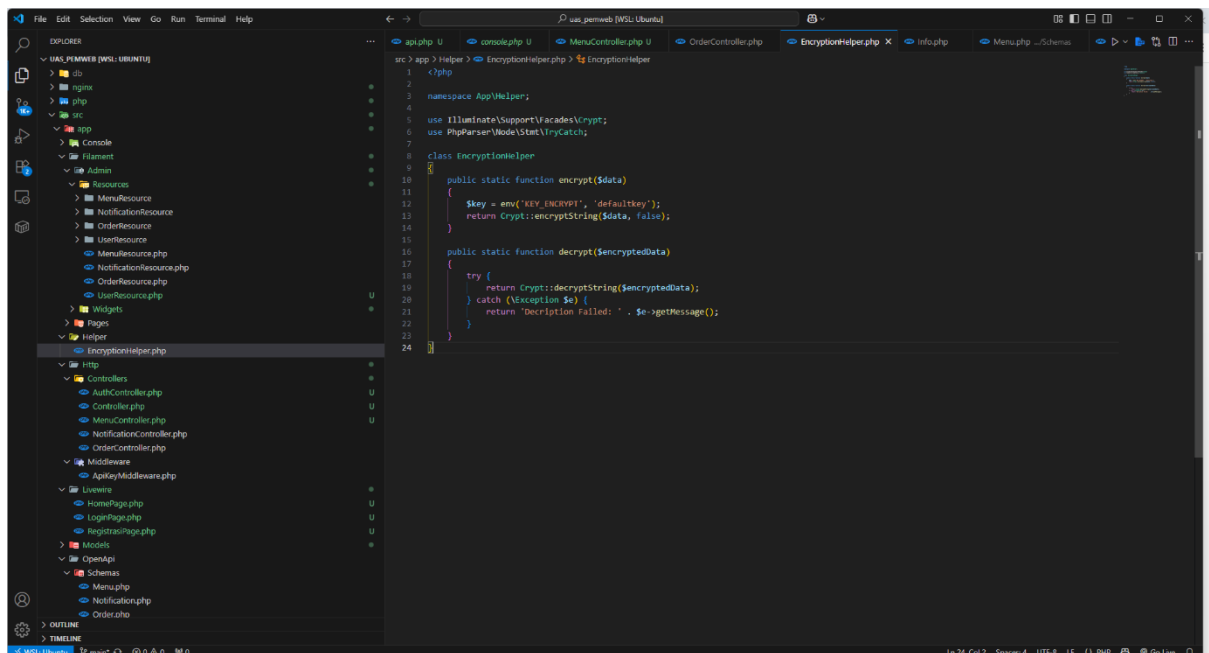
use Illuminate\Support\Facades\Route;
use App\Http\Controllers\MenuController;
use App\Http\Controllers\OrderController;
use App\Http\Controllers\NotificationController;

// Untuk Menu
Route::prefix('menus')->middleware('apikey')->group(function () {
    Route::get('/', [MenuController::class, 'index']);
    Route::post('/decrypt', [MenuController::class, 'decryptResponse']);
    Route::get('{id}', [MenuController::class, 'show']);
    Route::post('/', [MenuController::class, 'store']);
    Route::put('{id}', [MenuController::class, 'update']);
    Route::delete('{id}', [MenuController::class, 'destroy']);
});

// Untuk Order (dibuat sama persis strukturnya)
Route::prefix('orders')->middleware('apikey')->group(function () {
    Route::get('/', [OrderController::class, 'index']);
    Route::post('/decrypt', [OrderController::class, 'decryptResponse']);
    Route::get('{id}', [OrderController::class, 'show']);
    Route::post('/', [OrderController::class, 'store']);
    Route::put('{id}', [OrderController::class, 'update']);
    Route::delete('{id}', [OrderController::class, 'destroy']);
});

// Untuk Notification (dibuat sama persis strukturnya)
Route::prefix('notifications')->middleware('apikey')->group(function () {
    Route::get('/', [NotificationController::class, 'index']);
    Route::post('/decrypt', [NotificationController::class, 'decryptResponse']);
    Route::get('{id}', [NotificationController::class, 'show']);
    Route::post('/', [NotificationController::class, 'store']);
    Route::put('{id}', [NotificationController::class, 'update']);
    Route::delete('{id}', [NotificationController::class, 'destroy']);
});
```

5. Pembuatan Sistem Enkripsi



```
1 <?php
2
3 namespace App\Helpers;
4
5 use Illuminate\Support\Facades\Crypt;
6 use PhpParser\Node\Stmt\TryCatch;
7
8 class EncryptionHelper
9 {
10     public static function encrypt($data)
11     {
12         $key = env('KEY_ENCRYPT', 'defaultkey');
13         return Crypt::encryptString($data, false);
14     }
15
16     public static function decrypt($encryptedData)
17     {
18         try {
19             return Crypt::decryptString($encryptedData);
20         } catch (\Exception $e) {
21             return 'Decryption failed: ' . $e->getMessage();
22         }
23     }
24 }
```

6. Pengujian Api

Request URL

https://uas_pemob.test/api/noms

Server response

Code

Details

200

Response body

```
{
  "data": {
    "message": "success",
    "data": {
      "id": 1,
      "nama_barang": "Nasi Goreng Spesial",
      "kategori": "Makanan",
      "harga": 25000,
      "created_at": "2024-01-01T12:00:00Z",
      "updated_at": "2024-01-01T12:00:00Z"
    }
  }
}
```

Response headers

```
access-control-allow-origin: *
cache-control: no-cache, private
connection: keep-alive
content-type: application/json
date: Mon, 21 Jul 2025 20:01:46 GMT
server: nginx/1.28.0
transfer-encoding: chunked
x-powered-by: PHP/8.3.23
```

Responses

Code

Description

Links

200

Successful operation

No links

Media type

application/json

Controls Accept header

Example Value | Schema

```
{
  "message": "success",
  "data": {
    "id": 1,
    "nama_barang": "Nasi Goreng Spesial",
    "kategori": "Makanan",
    "harga": 25000,
    "created_at": "2024-01-01T12:00:00Z",
    "updated_at": "2024-01-01T12:00:00Z"
  }
}
```

Notifications

Notifications

GET

/api/notifications

Get all notifications

Returns a list of all notifications.

Parameters

Cancel

No parameters

Execute

Clear

Responses

Curl

curl -X 'GET' \
 https://uas_pemob.test/api/notifications' \
 -H 'accept: application/json' \
 -H 'x-api-key: qjmded' \
 -H 'x-csrf-token: '

Request URL

https://uas_pemob.test/api/notifications

Server response

Code

Details

200

Response body

```
{
  "data": {
    "message": "success",
    "data": [
      {
        "id": 1,
        "nama_barang": "Nasi Goreng Spesial",
        "kategori": "Makanan",
        "harga": 25000,
        "created_at": "2024-01-01T12:00:00Z",
        "updated_at": "2024-01-01T12:00:00Z"
      }
    ]
  }
}
```

Response headers

```
access-control-allow-origin: *
cache-control: no-cache, private
connection: keep-alive
```

Orders

Orders

GET

/api/orders

Get all orders

Returns a list of all orders.

Parameters

No parameters

Cancel

Execute

Clear

Responses

Curl

```
curl -X 'GET' \
  https://uas_pemweb.test/api/orders' \
  -H 'accept: application/json' \
  -H 'X-API-KEY: djaambred' \
  -H 'X-CSRF-TOKEN: '
```

Request URL

https://uas_pemweb.test/api/orders

Server response

Code

Details

200

Response body

```
{
  "data": "eyJpdii161MxkBUe0LzVtdFp4Y3Z0MMp3SF30V2c9PSTsInZhhV1IjoiIcDF-qWV11N0p1DUYVaNoMc3SHVFcWYQVYvWdDcmFhTWQeHE8L3NSdzNpb1FOVEYydkh1M0M1M0YVcURkQjZlZk30dTI1V13dwQ31tZ0JQ2Ize3kxclRmL2DVRk0o18RE1w0ZF-eEQ3dXV1cTV2cm0pUEgxQkZ2VXVqQkN6bV14TMTcVkvVK0NMpM2Z5cW1RcThOU1R1MnEwTHA1QXB2MmpHT0JucMRjokoaYnEwzjB1VnR3hWBPaeEVYyJhZCk1YzVMT0U2aEd1b1Fw5jc3ck1qS930UN0YmJW0H1UbXZ1cmNk1h2VUk0eks2T0uyMjB0dGx0S50V21HTVZ2Mk0MkcrSE1MT3oxQjV0RkVaWVBYkdPRzFHY1A4TjZ0czNpcETaaV3aT0J-VGJlY2VqeV1XajNVZ3NlVU0xc1BaMk5HuzJxbXBrSmhRY1Z6Q1VRTWNSG9NLMRqCkx3b1srTHQ3akWzdhRqVH3JYkdtTUR3VH8GTUozkzBjZ2R5Qm1F53pcS0RkZDQ1S1sRlc2R1U1BFRkQzQk11Z01B1K3W04YmJvWkNp0BYZDZ0U3g1d0g1L1oyYntBZHBXNzBycyt6bzHERLz3IdrVLNMc5aD11b3Y6M0p0c0xQU45UhhVUFQTUSpek0911w1bmFj1Jo1YzQ2YHR1ZG1Y2YzRjYwMTF1YzR0b0h0T11Y31Y1B1Zm9SRz1jMTUxZDh1OTQ1ZDgWjdazD0w0jhhNm04Yy1s1nRhzY161139"
```

Response headers

II. User Guide / Panduan Pengguna

1. User registrasi

Registrasi Pengguna

Nama Lengkap

plo

Alamat Email

plo@admin.com

Kata Sandi

.....

Konfirmasi Kata Sandi

.....

Daftar

Sudah punya akun? Masuk

2. User Login Jika sudah pernah Registrasi

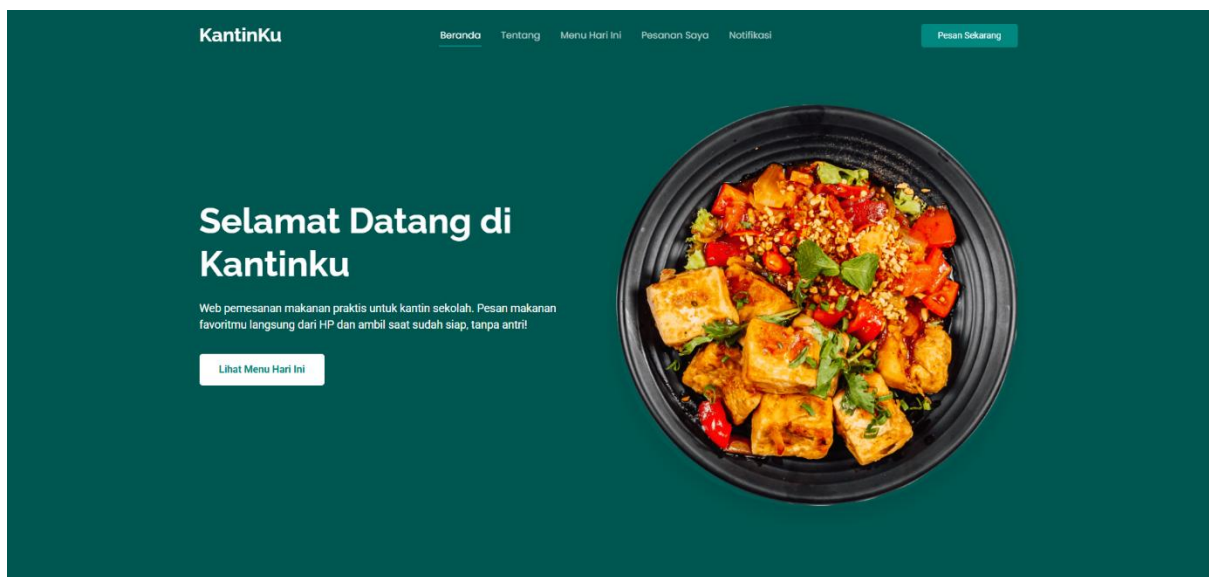
Login

Email

Password

Login

3. User Melihat halaman Dashboard



4. User Melihat halaman “Tentang”



10+

Tahun Melayani

Kantin Digital Sekolah yang Praktis & Higienis

Kami menyediakan platform pemesanan makanan yang cepat, mudah, dan higienis bagi para siswa dan guru di lingkungan sekolah.

Dengan sistem pemesanan online, pengguna dapat memilih menu harian, memesan makanan, dan mendapatkan notifikasi saat makanan siap diambil tanpa harus antri panjang.



Menu Bervariasi

Beragam pilihan makanan sehat dan lezat setiap hari.



Pemesanan Mudah

Pesan langsung dari web kapan saja sebelum jam istirahat.



Notifikasi Siap Ambil

Dapatkan pemberitahuan otomatis saat makanan sudah siap.



Higienis & Aman

Proses pengemasan makanan dilakukan sesuai standar kebersihan.

Lihat Menu Hari Ini

5. User Melihat daftar pilihan makanan yang tersedia

MENU HARI INI

Cek Pilihan Makanan



Nasi Goreng

Nasi goreng spesial dengan telur dan ayam.

Tersedia tanggal: 22 Jul 2025

Harga: Rp 15.000

Porsi sesuai standar

Order Sekarang →



Mie Ayam

Mie ayam lezat dengan kuah segar.

Tersedia tanggal: 22 Jul 2025

Harga: Rp 12.000

Porsi sesuai standar

Order Sekarang →



Es Teh Manis

Es teh manis dingin, pelepas dahaga.

Tersedia tanggal: 22 Jul 2025

Harga: Rp 5.000

Porsi sesuai standar

Order Sekarang →



roti bakar

rotibakar yang enak banget dah



rujak

enak banget rujak

6. User Order Makanan yang dia pilih

